Sahaj Bhadja

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Summary

Software engineer with 2+ years building **systems software**, **concurrent applications**, and **automated testing pipelines** in Python and C++. Strong foundation in algorithms, operating systems, and Linux environments with experience in CI/CD and cross-platform development.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Data Structures

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4.0*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Algorithms

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*
- Built **backend systems in Python** with fault-tolerant async processing, implementing job-state transitions and failure isolation that increased throughput by **3x** under concurrent workloads.
- Developed **data processing pipelines** handling 5K+ monthly jobs with validation logic, error handling, and CI/CD-backed releases achieving 99%+ reliability in production environments.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*
- Designed a **concurrent batch-processing engine using multithreading** (ThreadPoolExecutor), processing 1K+ files per batch with fault-tolerant execution and clear technical documentation.
- Wrote **clean, maintainable Python code** with modular architecture, participating in code reviews and collaborating with cross-functional teams in Agile sprints to implement new features.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*
- Built **end-to-end data pipelines in Python** processing 100K+ records using PyTorch, implementing algorithms for feature extraction with comprehensive logging and experiment tracking.
- Collaborated with **cross-functional research teams** to design software solutions, demonstrating clear documentation through flowcharts and well-structured code following engineering best practices.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*
- Engineered **cross-platform software with automated testing** using Jenkins, PyTest, and Mockito, implementing nightly build pipelines across Windows, Linux, and macOS with 20+ unit/integration test suites.
- Maintained **CI/CD pipelines with GitHub** for source control, building VM-based test automation that validated executables across 3 operating systems in Agile Scrum sprints.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*
- Developed **real-time synchronization systems** using WebSockets, implementing event-driven patterns for 2M+ records with microservice architecture and automated Jenkins deployments.
- Built **Docker-based packaging pipelines** with CI/CD automation, cutting deployment time by 50% while maintaining code quality through automated testing and staged rollouts.

Skills

Languages: Python, C++, C, Java, SQL, Bash, Shell scripting
Systems & Tools: Linux, Git, GitHub, Docker, CMake basics, Makefile basics, VS Code
Testing & CI/CD: PyTest, Mockito, Jenkins, GitHub Actions, automated testing, VM-based validation
Practices: Agile/Scrum, code reviews, technical documentation, cross-platform development

Projects

CPU-Scheduling Simulator Live | *C++, JavaScript, React, OS Algorithms*
- Implemented **CPU scheduling algorithms** (FCFS, SJF, SRT, Round Robin, MLQ) in C++, demonstrating deep understanding of OS internals, process management, and resource allocation with 25% TAT reduction.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed a **containerized Python application** with automated CI/CD pipelines, implementing validation checks and achieving production-grade reliability through systematic testing.

Chat2SQL Code | *Java, TypeScript, Docker, Maven*
- Built a **cross-platform application** working across Java backend and TypeScript frontend with Docker, Maven, and multi-environment development workflows on Windows, macOS, and Linux.